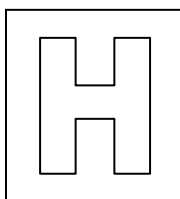


Candidate Name: _____

Class Adm No

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2011 Preliminary Examination II Pre-university 3

H2 Biology

9648/02

13 September 2011

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

This question paper consists of 22 printed pages.

[Turn over

Section A

Answer **all** questions in this section.

1.

- (a)** Plant cells have a cellulose cell wall outside the cell surface membrane. The cellulose microfibrils are embedded in a matrix of hemicellulose, pectin and proteins.

- (i)** Name the monosaccharide units that link together to form cellulose.

..... [1]

- (ii)** Name the precise type of bond between the monosaccharide units in cellulose and describe how it is formed. Draw an annotated diagram in the space below to illustrate.

name of bond

formation

.....

.....

.....

[4]

- (b) The cell surface membrane of a plant cell contains cellulose synthase enzymes. These enzymes make cellulose microfibrils. The synthesised microfibrils pass out through the enzymes as they leave the cell. This is shown in Fig. 1.1.

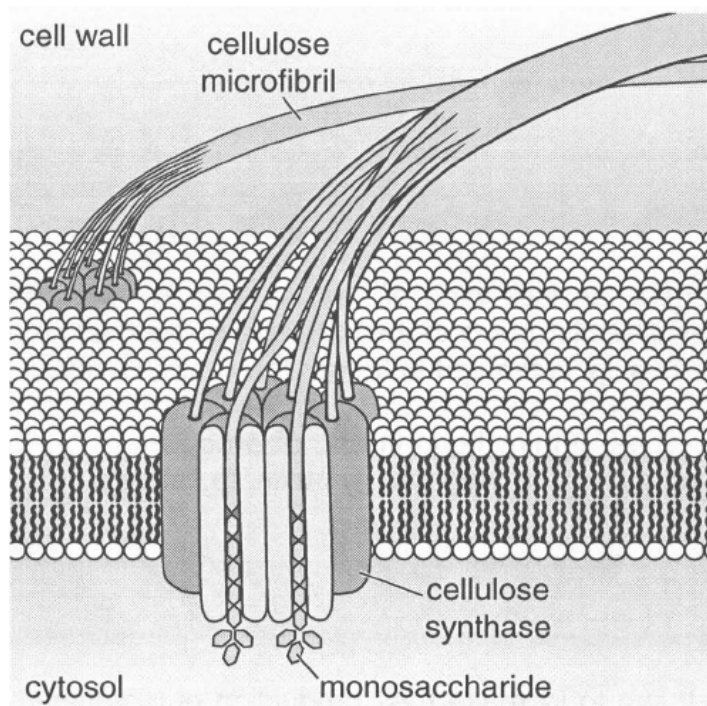


Fig. 1.1

- (i) Suggest why cellulose cannot be made outside the cell.

.....
 [1]

The cellulose fibres in the cell wall are made up of many cellulose microfibrils. Each microfibril is made up of many cellulose molecules lying parallel to each other.

- (ii) With reference to Fig. 1.1 and your knowledge of the structure of cellulose, explain how microfibrils are formed from cellulose molecules.

.....

 [4]

- (c) The growth of plant cells is caused by a rapid increase in the size of the vacuole and requires the loosening of the cellulose fibres in the cell wall.

- (i) Describe how water causes this increase in size of the vacuole to occur.

.....

.....

.....

..... [2]

- (ii) A fall in pH triggers the loosening of the cell wall by activating a protein called expansin. This causes disruption of the structure of the microfibrils. A plant growth substance, called auxin, is thought to trigger the change in pH.

From your knowledge of mitochondria and chloroplasts, suggest what process in the cell surface membrane is triggered by auxin, to cause a change in pH.

.....

..... [1]

[Total: 13]

2.

- (a) Describe how DNA is packaged in eukaryotic cells.

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..... [5]

- (b) The ends of each DNA molecule are lengths of DNA called telomeres. In humans, telomeres are made up of the base sequence TTAGGG, which is repeated up to 400 times.

In some cells, each time a DNA molecule is replicated, the last 150 - 200 bases are not copied. The telomeres therefore get shorter with each DNA replication until they become too short for DNA replication to take place.

Calculate the **maximum** number of replications that can occur with the DNA that contains 400 TTAGGG repeats.

Show your working.

..... [3]

- (c) In the body, only cells such as embryonic stem cells that need to divide regularly express telomerase. Other somatic cells do not normally express telomerase.

Most cancer cells which are continually dividing contain an enzyme, telomerase, which maintains the length of the telomere.

Suggest how these cancers could be treated.

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..... [2]

- (d) The enzyme that catalyses the replication of DNA checks for errors in the process and corrects them. This makes sure that the cells produced in mitosis are genetically identical.

Explain why checking for errors and correcting them is necessary.

.....

.....

..... [2]

[Total: 12]

3. Thyroid hormone, thyroxine, is produced by the thyroid gland, a gland in the throat region. Fig. 3.1 summarises the production and action of thyroxine in the body.

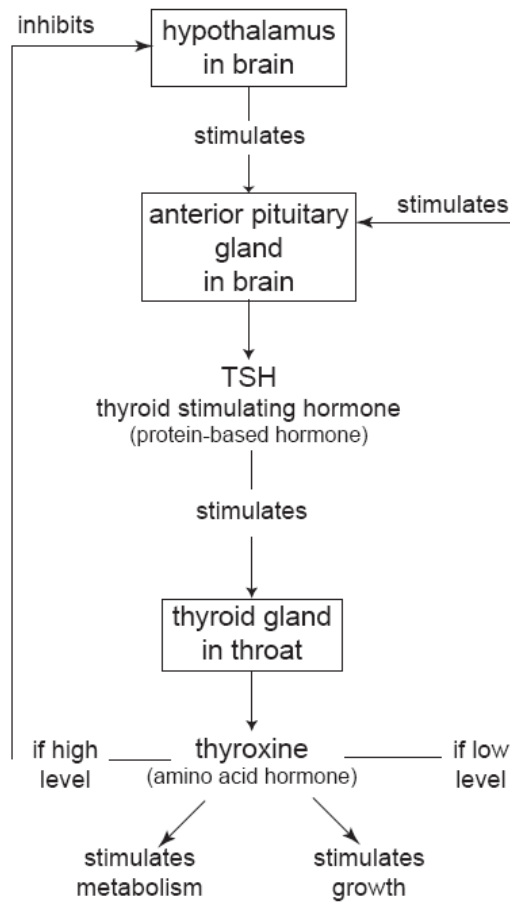


Fig. 3.1

- (a) Suggest how TSH travels to the thyroid gland.

..... [1]

The system shown in Fig. 3.1 is called a negative feedback system.

- (b) Describe a principle of a negative feedback system.

.....

 [1]

Fig. 3.2 shows general models for signal reception and transduction

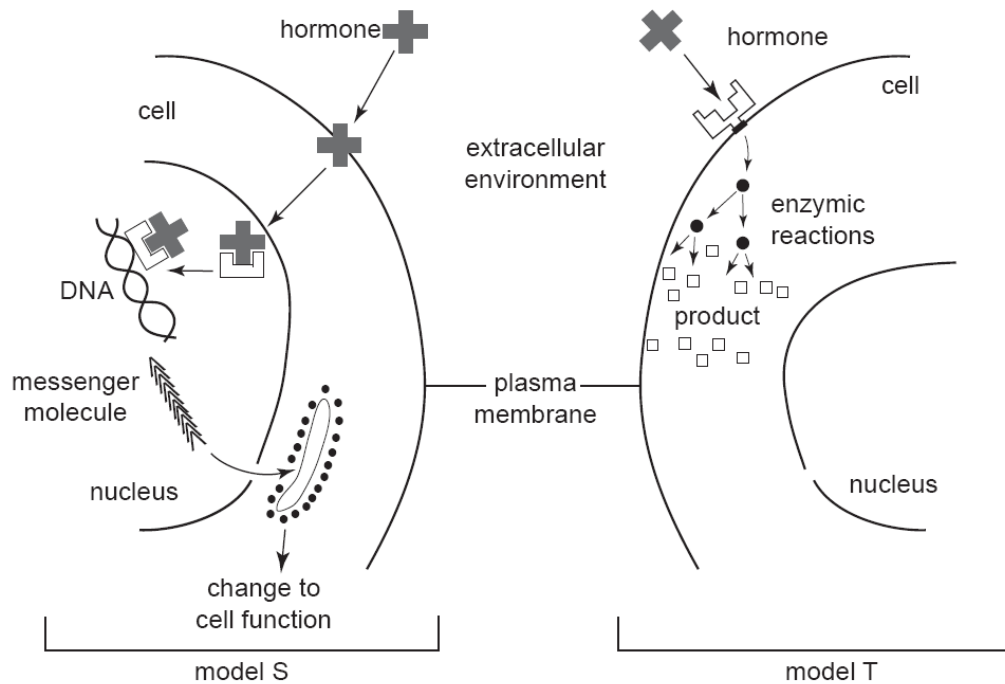


Fig. 3.2

(c) With reference to Fig. 3.2 and Fig. 3.1,

(i) which model, model S or model T, best corresponds to the mechanism of TSH?

..... [1]

(ii) explain your answer in (i).

.....

 [1]

(d) Testes and brain tissue do not respond to thyroxine. Suggest why.

.....
 [1]

The Fig. 3.3 shows some detail of a cell from a thyroid gland.

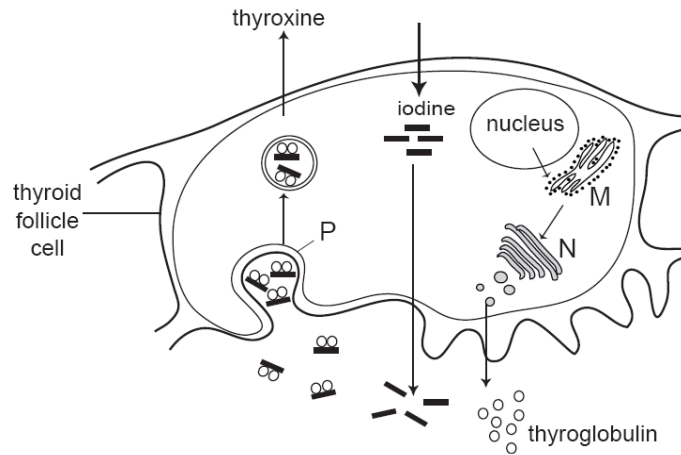


Fig. 3.3

(e) Outline the activity occurring in structure M.

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..... [4]

[Total: 9]

4. Liver cells damaged by hepatitis infection switch on a gene called Fas, which causes them to selfdestruct. Pioneering research has produced a strikingly successful treatment for hepatitis in mice. The Fas gene was silenced by the technique of RNA interference.

RNA molecules, 21 to 23 nucleotides long, were injected into mice with hepatitis. The sequence of this 'small interfering RNA' (siRNA) matched part of the Fas gene. Once in the liver cell the two strands of the siRNA were separated so that one strand could bind to the mRNA transcript of the Fas gene. This caused the mRNA to be destroyed by enzymes, therefore preventing the gene product from being made.

This therapy prevented liver cell death and considerably increased the survival of mice with hepatitis.

(a)

- (i) Describe a way in which the function of mRNA differs from that of DNA.

.....

 [1]

- (ii) Describe two ways in which the structure of siRNA differs from mRNA.

1

 2
 [2]

The technique of RNA interference has also been used to slow replication of HIV (Human Immunodeficiency Virus) *in vitro*. siRNA sequences that match the RNA genome of HIV can be used to trigger destruction of this RNA, preventing HIV from multiplying.

Another approach is to use RNA interference to silence genes for cell surface receptors, such as the CD4 and CCR5 molecules on human white blood cells. If these genes do not produce their protein antigens, HIV cannot bind to and infect the white blood cells.

- (b) Table 4.1 summarises some information about the two cell surface receptors used by HIV to bind to and infect white blood cells.

	cell surface receptor	
	CD4	CCR5
type of cell with this receptor	T lymphocyte white blood cells which divide by mitosis	macrophage cells which are long-lived and do not undergo mitosis
function of receptor	important roles in the immune system	limited, since 1% of people lack this receptor and show some resistance to HIV

Table 4.1

Experiments have been carried out where,

- siRNAs matching the CD4 mRNA were introduced into test tube populations of T lymphocytes;
- siRNAs matching the CCR5 mRNA were introduced into test tube populations of macrophages.

In both cases HIV was present but the presence of the siRNAs reduced its replication.

- (i) Use Table 4.1 to suggest with reasons which of the two test tube experiments showed most reduction of HIV replication.

.....

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..... [2]

- (ii) Explain which receptor would be the best target for RNA interference if the approach was used as a therapy for humans infected with HIV.

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..... [1]

- (c) Outline the process which occurs after HIV binds to its host.

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..... [5]

[Total:11]

5. Four different eye pigments in the fruit fly, *Drosophila melanogaster*, are made from the amino acid tryptophan. A simplified metabolic pathway of pigment production is shown in Fig. 5.1.

Three different gene loci control the pathway. Each locus has two alleles. These alleles are **V** or **v**, **C** or **c** and **B** or **b**, as shown in Fig. 5.1.

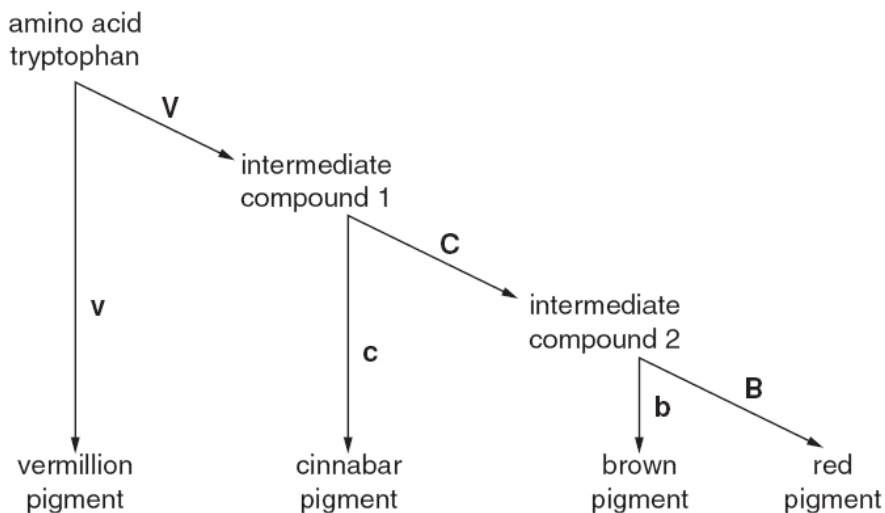


Fig. 5.1

(a)

- (i) Using the information in Fig. 5.1, deduce the phenotypes of flies with the following genotypes:

<i>genotype</i>	<i>phenotype</i>
VvCcBb	
vvCCBB	
VvccBB	

[3]

- (ii) State the term that is applied to this type of gene interaction.

..... [1]

- (iii) Explain how the products **coded for** by the genes interact to give the different pigments.

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.....

..... [3]

- (b) A mutation in another gene at another locus in *Drosophila* gives rise to white-eyed flies. The red eye allele of this gene (**R**) is known to be dominant to the white eye allele (**r**).

A student crossed a red-eyed fly with a white-eyed fly, expecting to get an F₁ generation of red-eyed flies. In fact, the results were as shown in Table 5.2.

phenotype of fly	number of offspring
red-eyed female	27
red-eyed male	0
white-eyed female	0
white-eyed male	23

Table 5.2

- (i) The student first suggested that the reason for there being red-eyed and white-eyed flies in the offspring was that the red-eyed parent was heterozygous.

Explain why this **cannot** be the correct explanation for the results shown in Table 5.2.

.....

 [2]

- (ii) In *Drosophila*, the males are the heterogametic sex, possessing two different sex chromosomes, X and Y.

Draw a genetic diagram to show how the results shown in Table 5.2 could have been produced.

[3]

(iii) The chi-squared (χ^2) test can be used to analyse the results in Table 5.1.

The expected ratio of red-eyed females to white-eyed males is 1:1.

Use Table 5.2 to calculate a value for chi-squared (χ^2).

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad df = n - 1$$

Key to symbols:

Σ = 'sum of ...'

df = degrees of freedom

n = number of classes

O = observed value

E = expected value

Table 5.3

phenotype of fly	O	E	O - E	(O - E) ²	$\frac{(O - E)^2}{E}$
red-eyed female					
white-eyed male					

$$\chi^2 = \dots\dots\dots$$

Use your calculated value of χ^2 and the table of probabilities shown in Table 5.4 to test the significance of the difference between the observed and expected results.

degrees of freedom	probability, p			
	0.90	0.50	0.10	0.05
1	0.02	0.45	2.71	3.84
2	0.21	1.39	4.61	5.99

Table 5.4

State your conclusion below

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..... [4]

[Total:16]

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6. An experiment was conducted to examine the effects of glucose and lactose on the levels of β -galactosidase in *E.coli*. Lactose and glucose were added to a culture of bacteria at the start of the experiment and the levels of each were measured at specific time intervals.

The results are shown in Fig. 6.1 below.

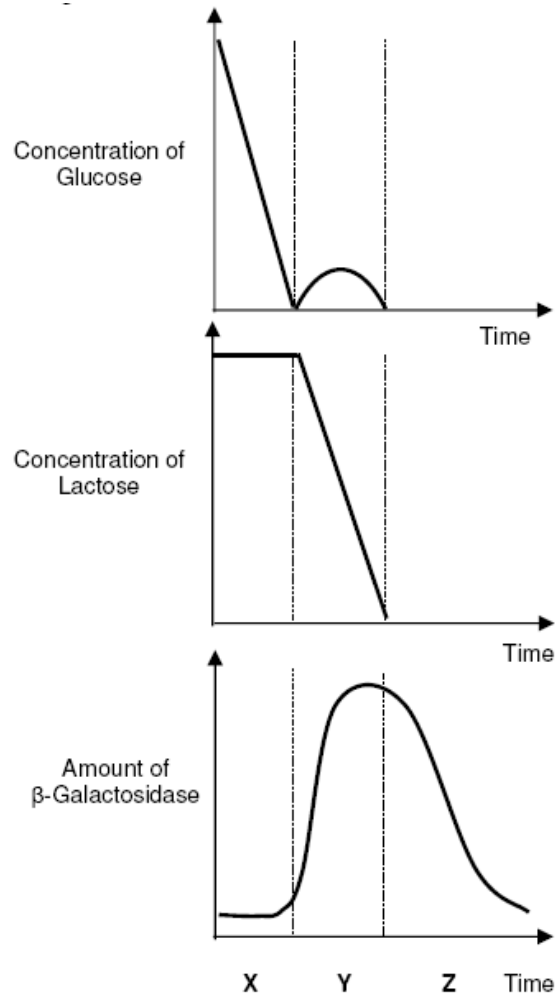


Fig. 6.1

- (a) State the two mechanisms of regulation of lac operon.

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..... [2]

- (b) With reference to Fig. 6.1, explain how expression levels of β -galactosidase is regulated by the lac repressor in periods X and Y.

X
.....
.....
..... [2]

Y
.....
.....
..... [2]

[Total:6]

7. The domestic dog, *Canis familiaris*, is one of the most striking examples of variation in a domestic animal because of the large number of different breeds. It was first domesticated at least 14 000 years ago and is thought to have descended from the grey wolf, *Canis lupus*.

- (a) Table 7.1 shows the classification of the grey wolf. Complete the table by writing your answers in the appropriate boxes.

kingdom	
	Chordata
	Mammalia
	Carnivora
	Canidae
genus	
species	

Table 7.1

[2]

- (b) Define the term species.

.....

.....

..... [2]

- (c) Five ancient breeds of dog are recognised, from which all other breeds are thought to have descended by artificial selection. One of these ancient breeds is the greyhound, *Canis familiaris leineri*. Most greyhounds are bred for racing.

Suggest how artificial selection in greyhounds has taken place.

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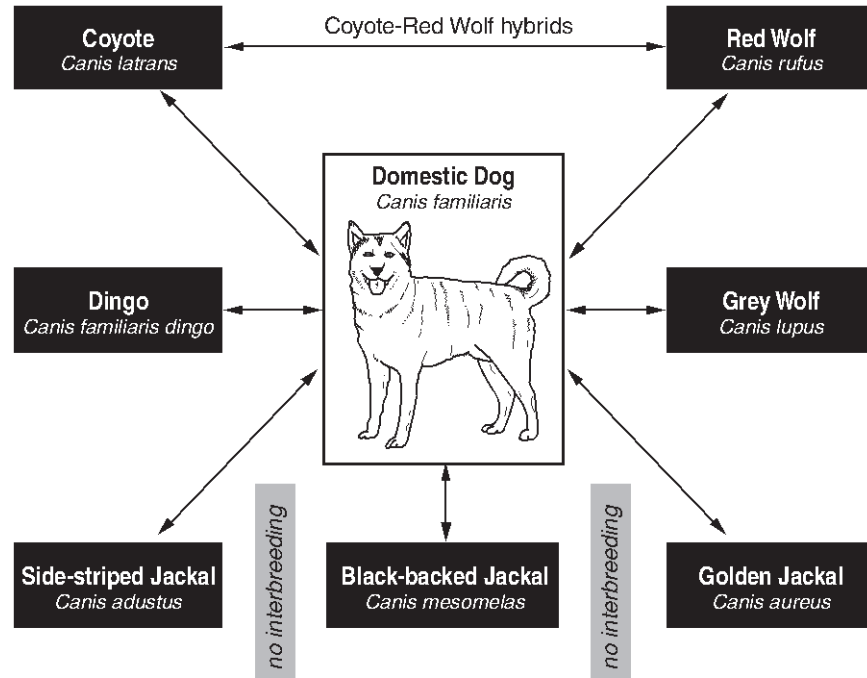
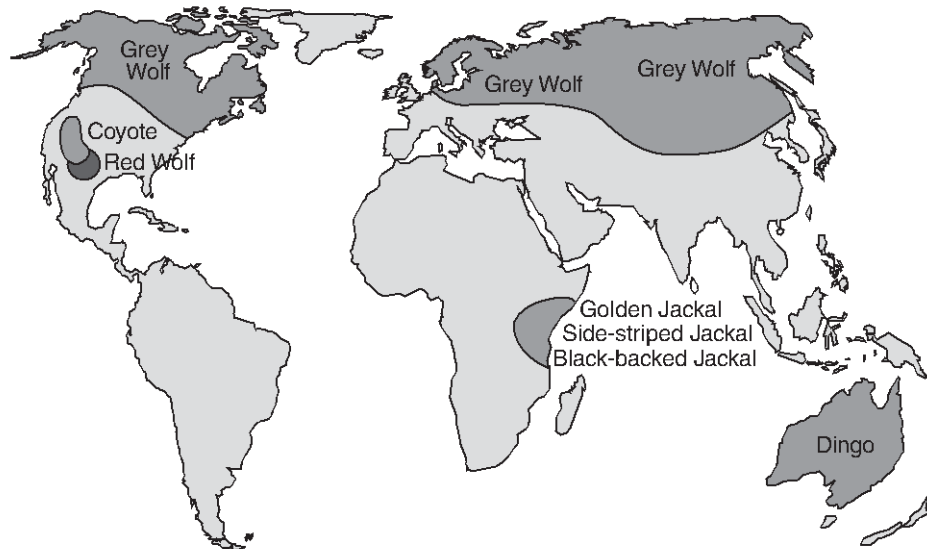
.....

..... [4]

- (d) The global distribution of most of the species belonging to the genus *Canis* is shown in Fig. 7.2.

As a result of the spread of human culture, distribution of the domestic dog is global. The domestic dog has been able to breed with all other members of the genus to form fertile hybrids.

The ranges of the three distinct species of jackal overlap in the Serengeti of East Africa.



Source: Advanced Biology A2 (2008). Page 189, BIOZONE International. By Tracey Greenwood, et al.

Fig. 7.2

- (i) State the type of isolating mechanism preventing, the three species of jackal from interbreeding

..... [1]

- (ii) State the type of isolating mechanism preventing, the dingo mating with all other members of the genus *Canis* apart from the domestic dog.

..... [1]

- (iii) Using the information in Fig. 7.2, explain whether members of the genus *Canis* should be described as separate species.

.....

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.....

.....

..... [3]

[Total:13]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

8.

- (a)** Outline how the resting potential is maintained. [6]
- (b)** Describe the transmission of an action potential along a motor neurone. [8]
- (c)** Discuss the roles of synapses in the nervous system. [6]

[Total: 20]

9.

- (a)** Describe the main features of the Calvin Cycle. [8]
- (b)** Outline the role of NADP in photosynthesis. [6]
- (c)** Explain how ATP is synthesised using light energy in photosynthesis [6]

[Total: 20]